



Human System Integration 101

Dr. Dimitri Masson

incose.org



HSI 101.

A brief introduction to HSI



Dimitri Masson

Co-chair – Sector 2

Assistant Professor in HSI
Incase TLI cohort 8



Grace Kennedy

Chair – Sector 4

Principal Consultant HF & SE @ Acmena

INCOSE HSI Wg



Firstname Lastname

Co Chair – Sector 1

Chief Technologist
Autonomous Systems
@ Collins Aerospace



Firstname Lastname

Co-chair – Sector 1

Flight Deck Lead Engineer
@ Boeing



Firstname Lastname

Co-chair – Sector 2

CEO, @ end to end Systems



Firstname Lastname

Co-chair – Sector 3 Asia-
Oceania)

COULD THIS BE YOU?

HSI Relevance

- Vision 2035
 - “Approaching 2035, socio-technical systems will be increasingly **autonomous, incorporating more AI, will be massively interconnected,** and must be collaborative, safe, secure and trusted.
 - Analyzing and predicting system behavior will become more challenging, but systems engineers will be expected to **analyze, design, and evaluate these systems with human and natural principles in mind**. Systems engineering practice **will include HSI methods for evaluating human factors and usability, identifying emergent behaviors, and detecting and managing unanticipated behaviors**.
 - Improvements in HSI methods, **human behavioral simulation** and **human-in-the-loop simulation** capabilities will help power the next generation of systems engineering practices and tools, allowing the systems engineer to better understand possible non-deterministic outcomes and cope with uncertainty.
 - **Research in HSI will be infused into systems engineering practice and become a core competency of the systems engineer.”**
- IS2026
 - Best Paper : **246 - Human-in-the-Loop Safety for ADAS: Blending Kansei, HSI, and AI to Reduce False-Positive Interventions**

Starter for 10...

Q1. What does HSI
stand for?

(in terms of today's talk)

- A. Human Systems
International
- B. Human Systems Interaction
- C. Human Systems Integration
- D. Human Systems
Intersection

You can type your answer in the chat

Starter for 10...

A. What does HSI stand for?

A. Human Systems International

B. Human Systems Interaction

C. Human Systems Integration

D. Human Systems Intersection

Q2. Who does HSI?

For 500 points

- A. Systems Engineering People
- B. Human Factors Practitioners People
- C. Human Computer Interaction People
- D. Operational Research People

You can type your answer in the chat

For 500 points

A. Who does HSI?

A. Systems Engineering
People

B. Human Factors People

C. Human Computer
Interaction People

D. Operational Research
People

TRICK QUESTION!

It's all of the above.

Q3. What is being integrated?

**The Finale... for
1000 points**

- A. Humans and Technological Systems
- B. Humans into Systems Design
- C. HF and SE Approaches
- D. Teams of HF and SE

You can type your answer in the chat

The Finale... for 1000 points

A. What is being
integrated?

A. Humans into Technological Systems

B. Humans into Systems Design

C. HF and SE Approaches

D. Teams of HF and SE

IT DEPENDS ON YOUR PERSPECTIVE

What is Human Systems Integration?

Also known as Human Factors Integration

Short Definition: “Human Systems Integration (HSI) is defined as an approach of systems engineering (SE) that integrates technology, organizations and people effectively.”

“HSI is the **transdisciplinary sociotechnical and management PROCESS** used to ensure that the **human elements** of a system are **appropriately addressed** and **integrated** within the wider **systems engineering lifecycle** and **management approach** to a project. HSI involves **analyzing, designing and evaluating technological, organizational and human systems**. “ INCOSE SE Handbook (5th Edition, 2023)

What it is

Who

Involving what?

What activities are being done?

Value Proposition for HSI



Performance

HSI activities support the development of systems to ensure:

- Meet the required capability
- Consider human possibilities, limitations and necessities
- Optimise the sociotechnical system performance

Risk

Manage program risk:

- Earlier in the lifecycle
- Reducing costs to rectify defects if they were discovered later

Control risks throughout the lifecycle:

- Safety during operations and maintenance
- Long term damage to health or wellbeing
- Loss of life

Cost

- Reduce program costs (see program risk)
- Reduce and optimise project costs over the whole systems lifecycle.

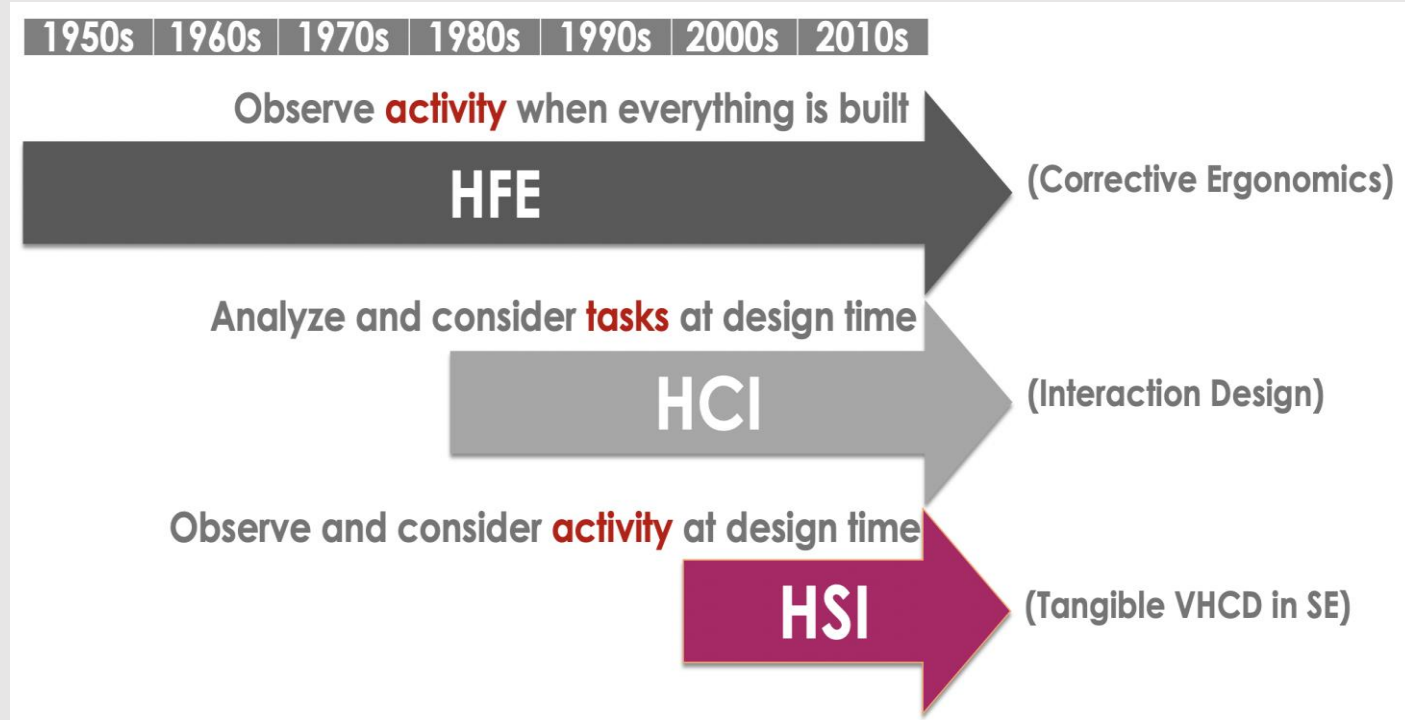
Time

- Reduce risk of programmatic time overruns
- Improve efficiency of effort and reduction of rework

Organisation

- Improve integration during projects, reducing silos
- Develops the skills of the SE and HF workforce
- Used across project portfolios

History of HSI



From Guy Andre Boy (2020). Human-Systems Integration: From Virtual to Tangible.

History of HSI

Pre-1980s

From...

- Technology-centred product design and development
- Focussed on training of personnel once the system development was almost complete.

Fitting the human to the technology



1980s

Then...

- HSI borne out of US Army efforts to formalise whole of system processes.
- MANPRINT (Manpower and Personnel Integration) program 1986 sought to address the integration of the human aspects into the SE activities
- Bringing in HF analyses earlier into the development
- Limited to the users of the operational system

Fitting the technology to human needs



Modern Day

Now

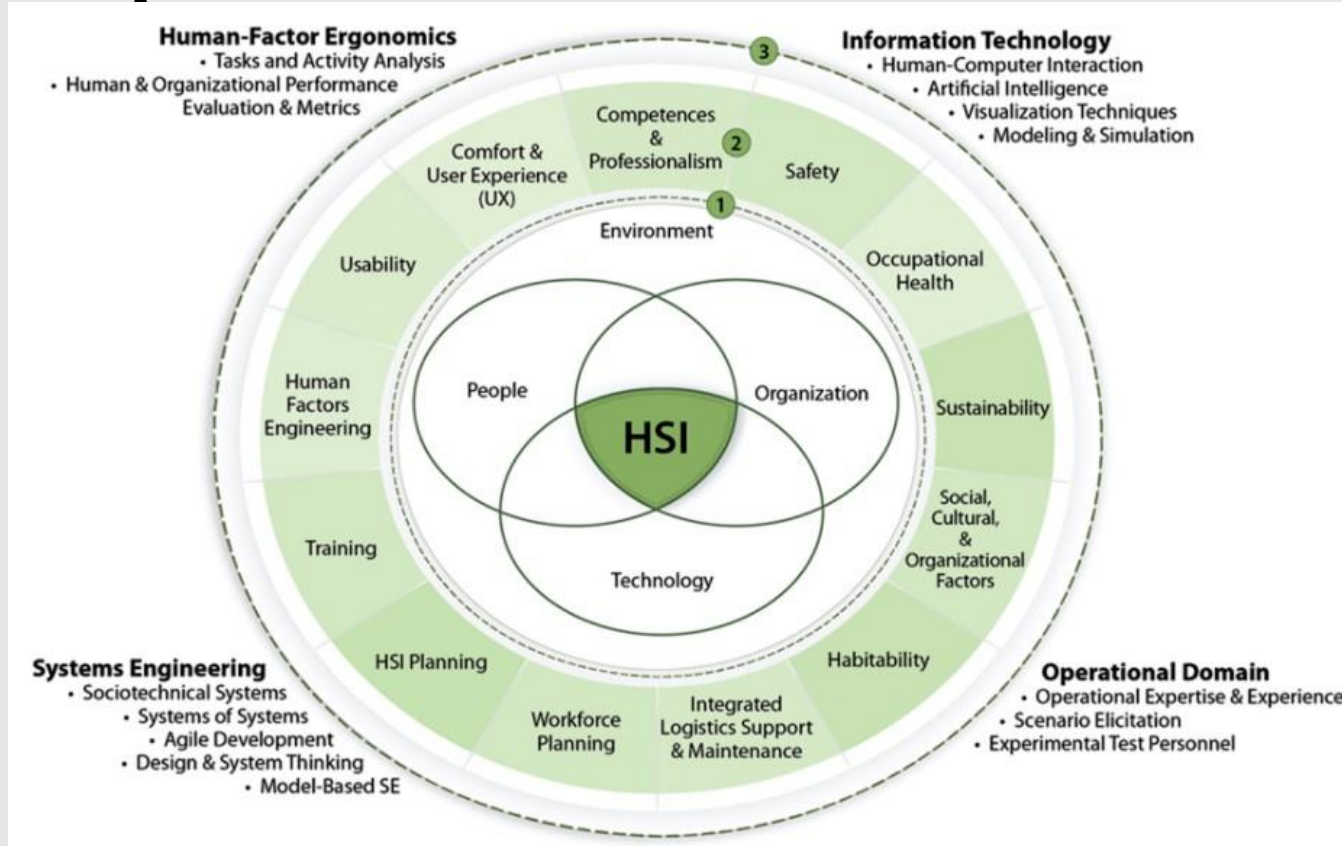
- Not just defence domains
- All humans and organisations at any stage of the system lifecycle
- Human-in-the-loop simulation in virtual engineering

Human-centred design that allows the human to “be human”

The Scope of HSI

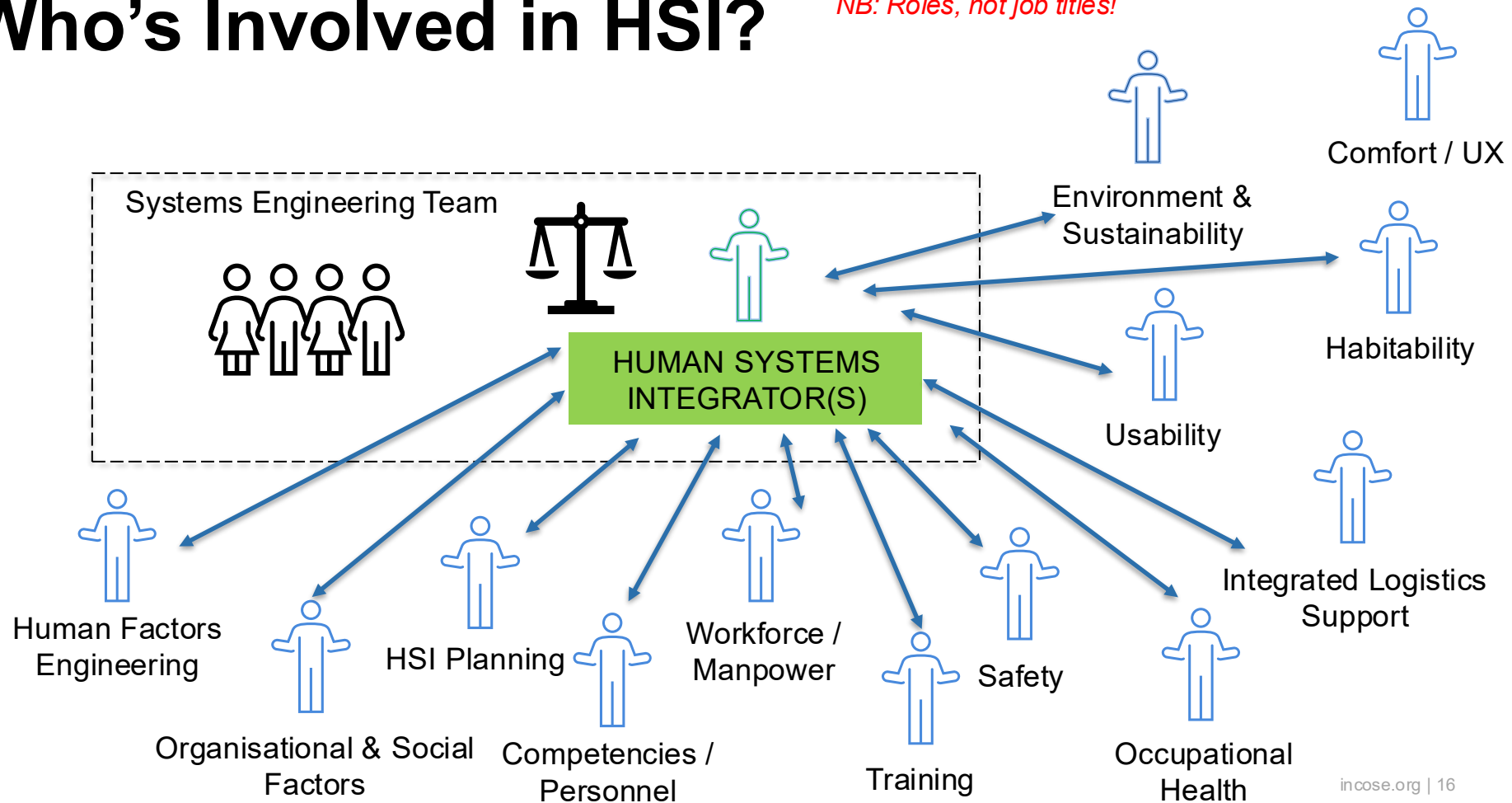
[Scope of HSI from the INCOSE Systems Engineering Handbook: a

guide for systems lifecycle processes and activities 2019]



Who's Involved in HSI?

NB: Roles, not job titles!



Where do HSI Practitioners Come From?

Niches of niches

Human Factors/Ergonomics

Social Sciences:
Psychology, Physical
Sciences, Cognitive
Sciences, OHS, etc



HF/E Professionals



*Not many specialist
tertiary degrees
in HF or SE*



*Experience doing HSI
or working in cross-
disciplinary teams*



Specialism in HSI

Systems Engineering

Engineering Disciplines:
Civil, Mechanical,
Electrical, etc



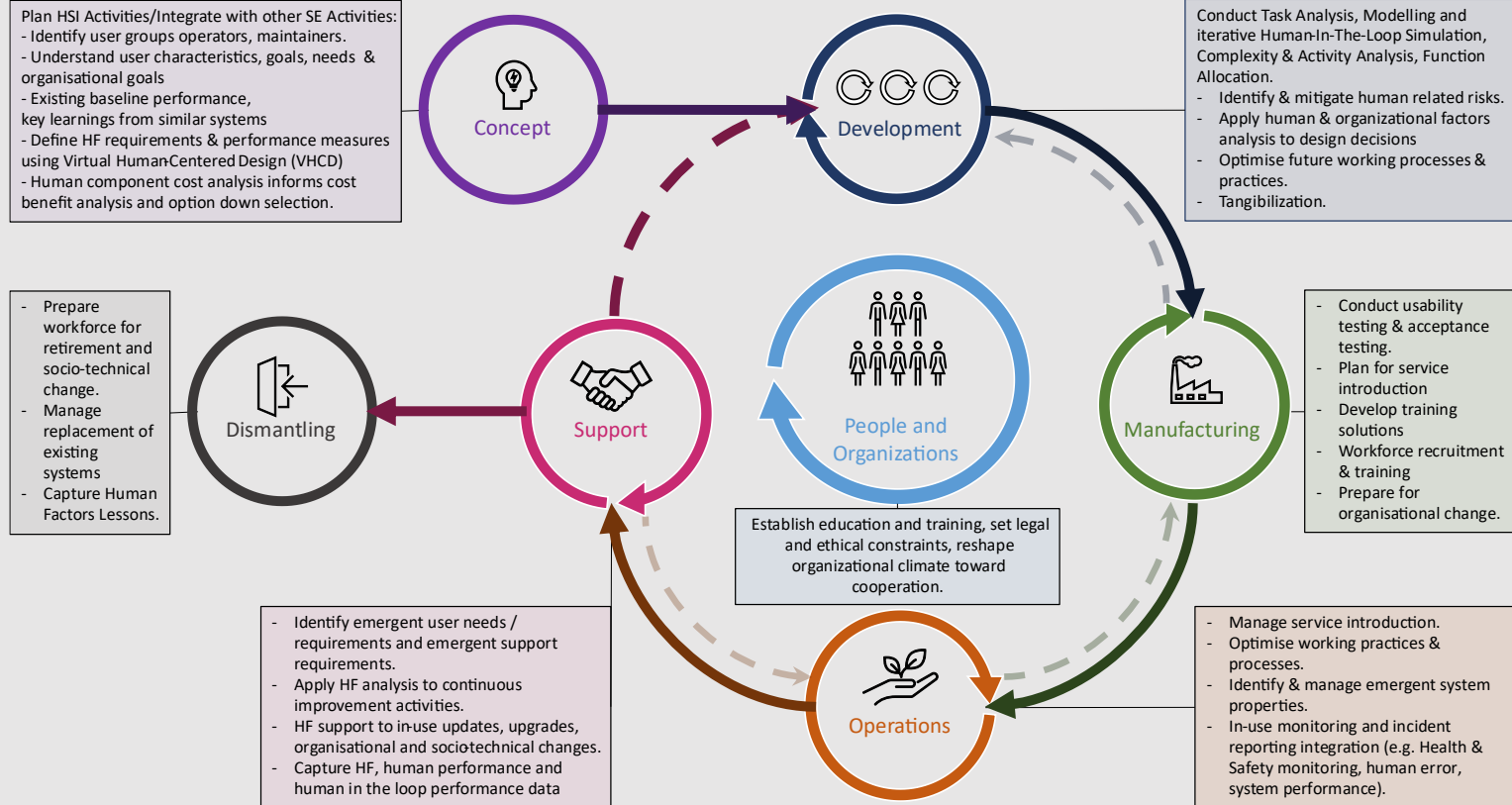
Systems Engineering
Practitioner



Non-traditional routes

Non-traditional routes

HSI Activities in the SE Lifecycle



Tailoring HSI

Not One Size Fits All

- Scale your HSI effort to:
 - Complexity of human interaction with the system
 - Novelty of the system and degree of automation
 - Level of risk and safety criticality
 - Degree of organisational change
 - Resources available and mandated standards

HSI must **coordinate** with SE activities throughout the lifecycle — not sit alongside them as an optional add-on.

HSI Examples

Unmanned Oil Rigs – Designing a Control Room

1:30 PM

1:30 PM - 1:55 PM JT
25 minutes

306 - Mitigating Retrofit-Oriented Bias in Industrial Controller and HMI Development through a Five-Phase Upstream Process

[Remove](#)

📍 Room 313/314 - 3rd Floor

This session introduces a five-phase upstream development process for industrial equipment to counter retrofit bias. By addressing structural factors like network externalities and path dependence, the process ensures...

2:00 PM

2:00 PM - 2:25 PM JT
25 minutes

165 - Involving Non-Technical Stakeholders in the System Adoption/Implementation Process: a Case-Study on the Cleaning Industry

[Remove](#)

📍 Room 313/314 - 3rd Floor

A successful system integration requires the acceptance of and adoption by the end-users and acquirer. This complex social system adds depth to design decisions made within a development process. Tailoring tools and...

2:30 PM

2:30 PM - 2:55 PM JT
25 minutes

65 - Beyond the Model: Human Factors in Successful Systems Engineering and Product Management

[Remove](#)

📍 Room 313/314 - 3rd Floor

Delivering complex software products takes more than technical rigor—it requires balancing systems engineering with human factors. This session shows why traditional SE and MBSE can falter when isolated, and offers...

3:00 PM

3:00 PM - 3:25 PM JT
25 minutes

311 - Formalizing Document Assurance: A Topological Framework for Verification, Validation, and Human Accountability

[Remove](#)

📍 Room 313/314 - 3rd Floor

Who is accountable when AI assists systems engineering documentation? Current V&V frameworks lack formal mechanisms to integrate structural compliance with fitness-for-purpose or attribute responsibility for quality...

Introduction

- Oil Rigs are remote locations, with harsh conditions for human operators.
- Design production facility that can be operated without human on site during a long period
- Robotic operations are a game changer
- The study aims at defining the emerging human functions when introducing robotics in operations.

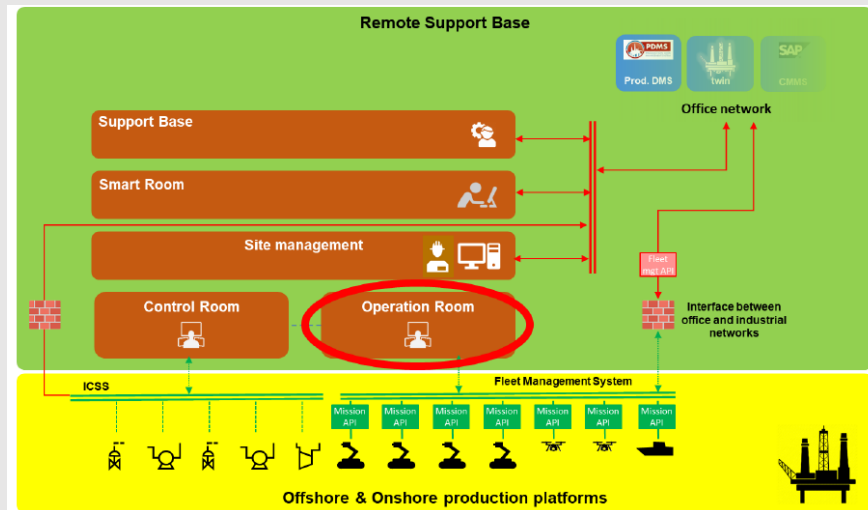
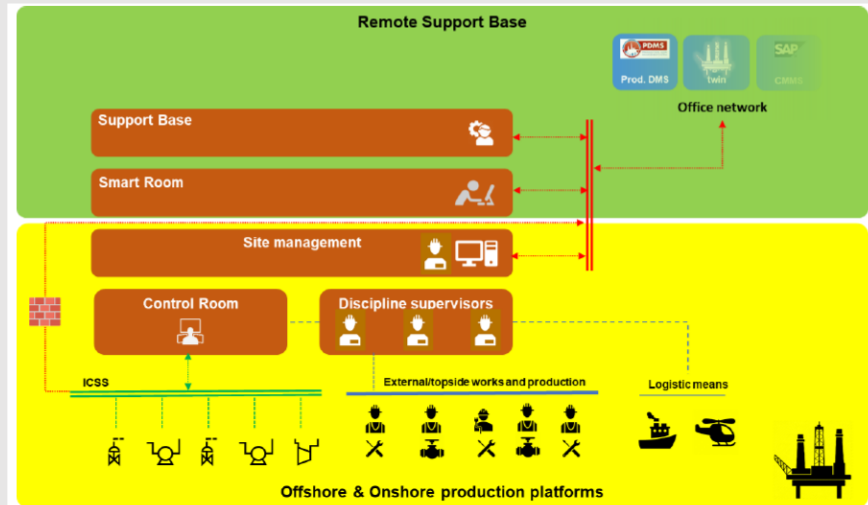


Redesign of the organization of the platforms

- Facilities operated by robots, which would be controlled remotely via the operating room.
- Operating room next to the existing control room, which would also become remote.
- No need for continuous human presence.
- From the operating room, remote supervision and coordination of the robots by the operators. The robots would autonomously perform the activities that field operators used to do on the installations: routine tasks, maintenance, emergency operations.
- Human interventions limited to short-term maintenance campaigns.

Objectives:

- To study the impact of the introduction of robotics in the management of operations, which will be carried out from the remote control room and operating room of the site.
- Specify the organization of the new "operating room" entity and identify the skills that the operators of this room must have.

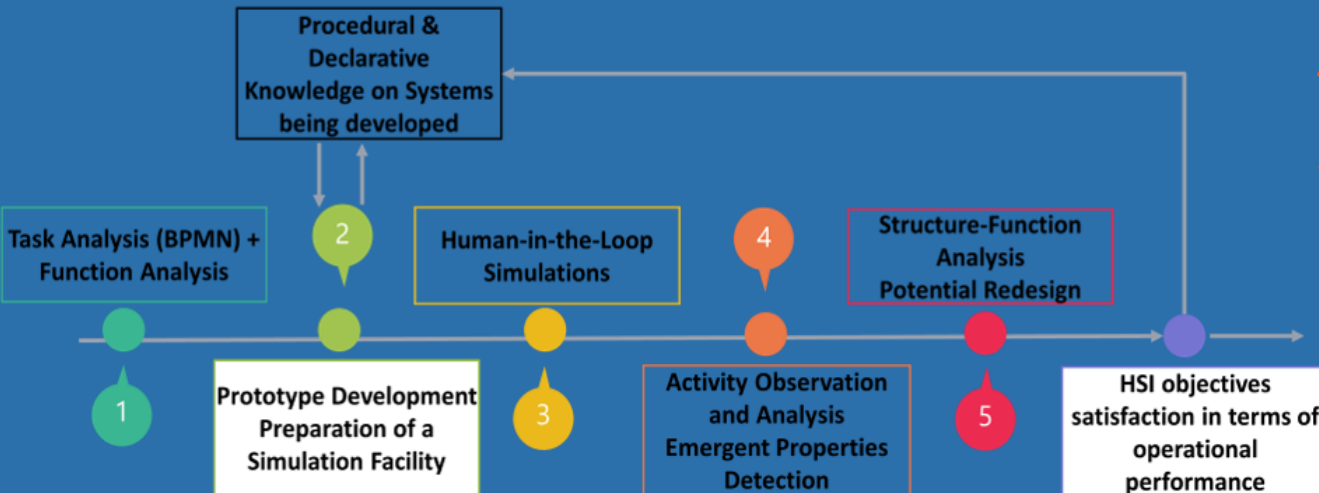


Project objectives

- Can you develop a control room for a fleet of robots before the robots exists ?
- What will be the impacts of controlling such a fleet on the overall organization ?
- Can we apply the PRODEC methodology from the MMT project to another case study ?



How do we go about



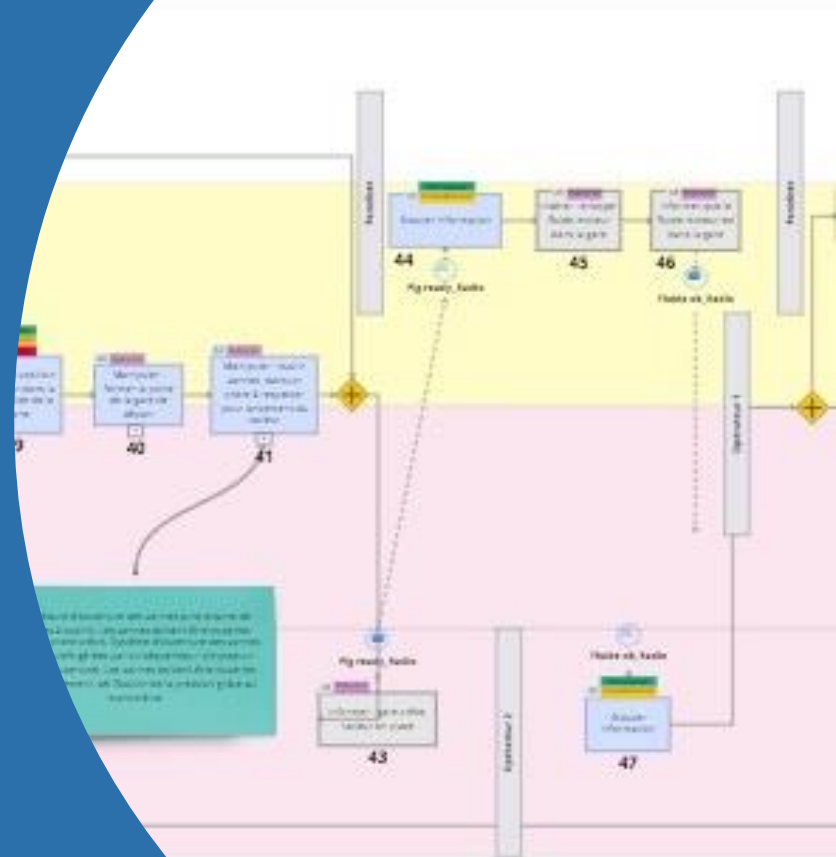
There is too many activities to cover them all

1. Select a pool of scenarios of increasing complexity.
2. Model one scenario
3. Identify necessary functions for each agents
4. Create one or several new allocation of these functions
5. Simulate these processes, analyse and add these to the pool of processes

Modelling

- BPMN Modelisation :
- Multi-agents modelling
- Multi-level modelling
- Cross validation with the experts

https://miro.com/app/board/uXjVMj0x6cc=



Functional Analysis

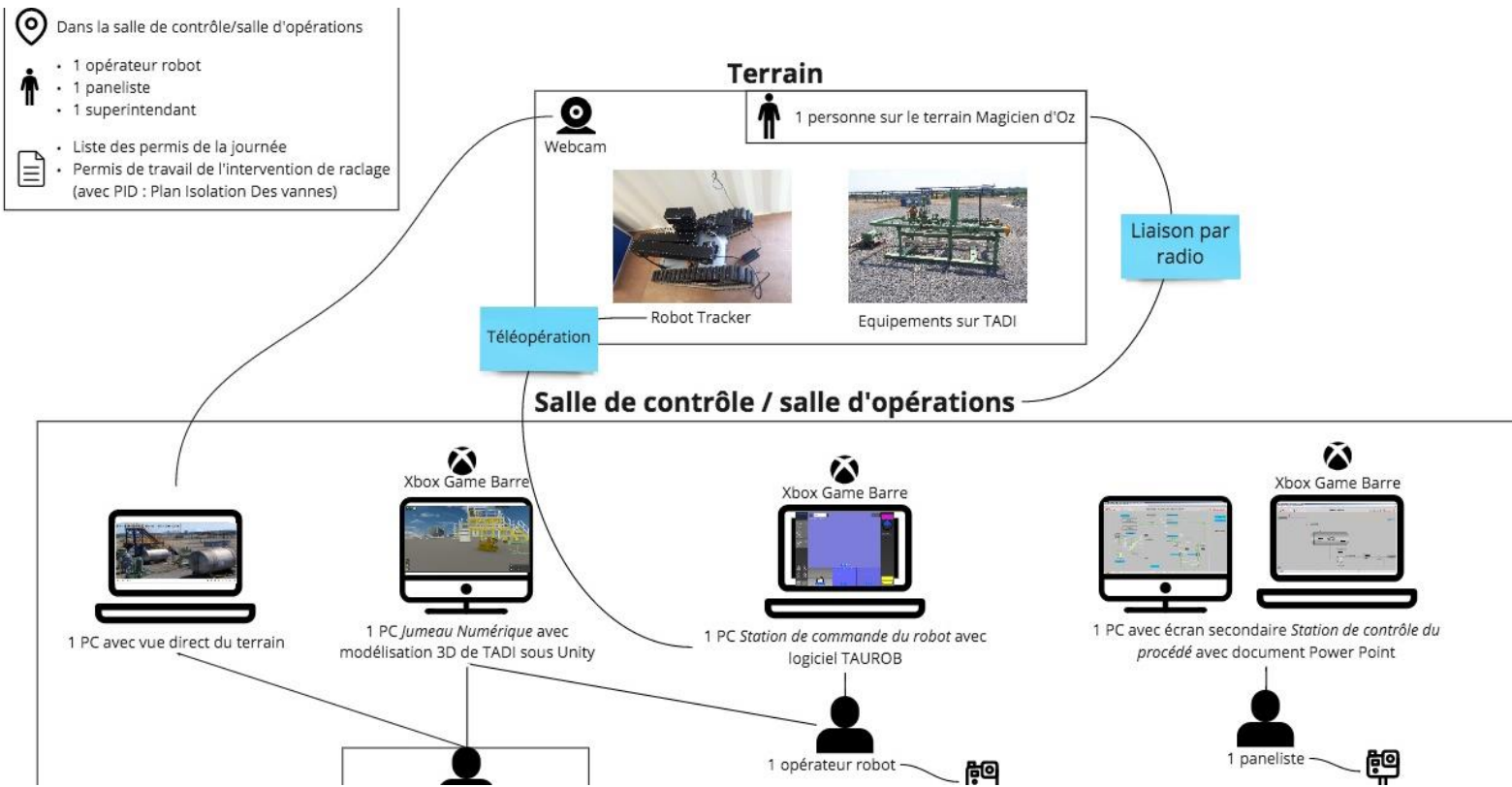
- Function are attributed to the different agents
- Ontological approach to generalize functions
- Classification according to
 - Situational Awareness Level (*Endsley, 1995*)
 1. Situation Awareness (S)
 2. Decision making (D)
 3. Action (A)
 - Rasmussen levels of functions (*Rasmussen, 1983*).
 1. Skills based (1)
 2. Rules based (2)
 3. Knowledge based (3)



New task allocations

- Based on As-is scenario
- Functional Analysis
 - e.g. verification and inspection task are kept on human
 - Listening and informing become receiving and sending information
- Predicted functional capabilities of the robots
 - We can predict from the analysis some capabilities (carrying, identifying etc..)
- Prescribed organisation of the oil rig (
 - separation control room (process) / operation room (robot fleet)
 - Iterative process involving experts and creative sessions
 - Several concurrent scenario have been proposed and discussed for the simulations

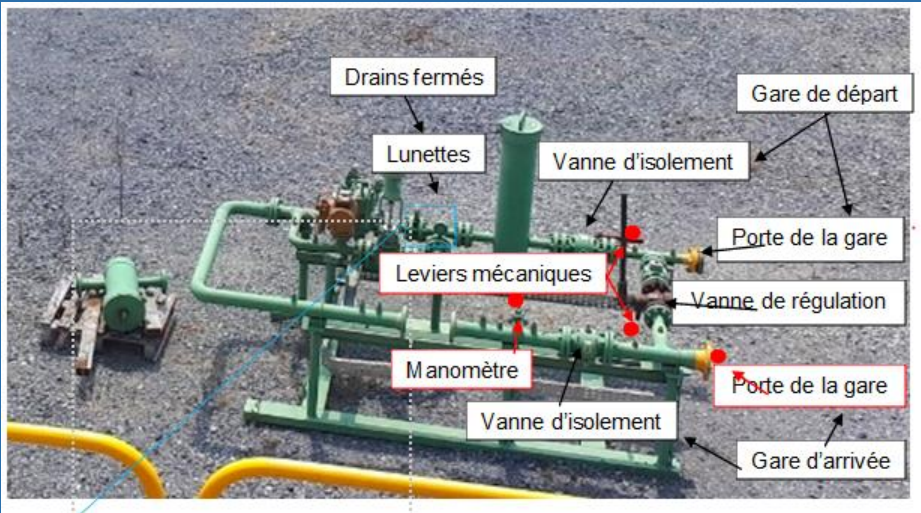




Human in the loop simulations Wizard of Oz

Human in the loop simulations

Physical and digital twin



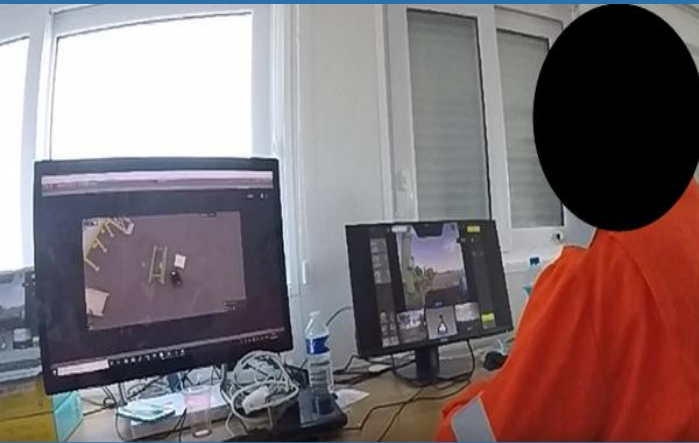
Fake equipment to simulate a pig launcher



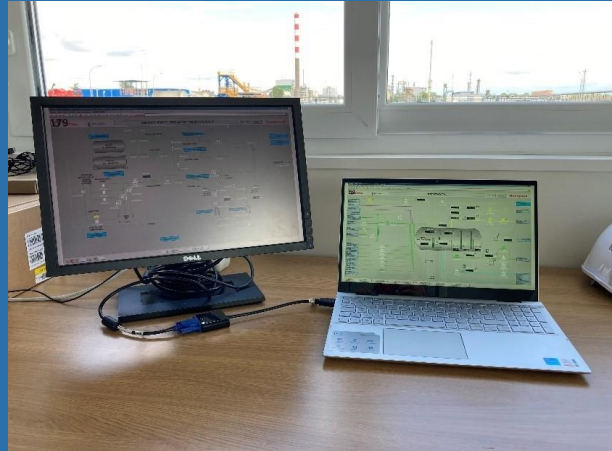
Digital twin of the equipment and robot

Human in the loop simulations

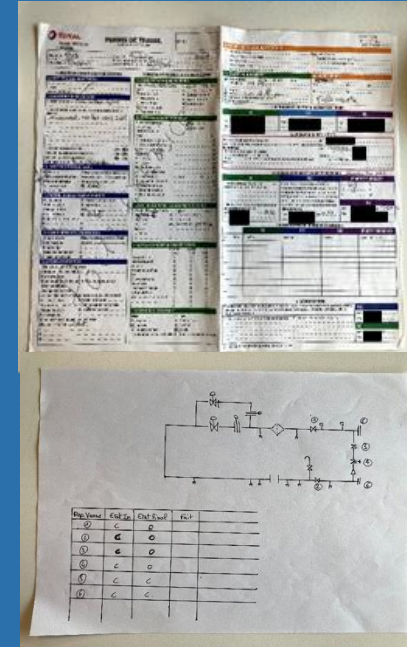
User interfaces and resources



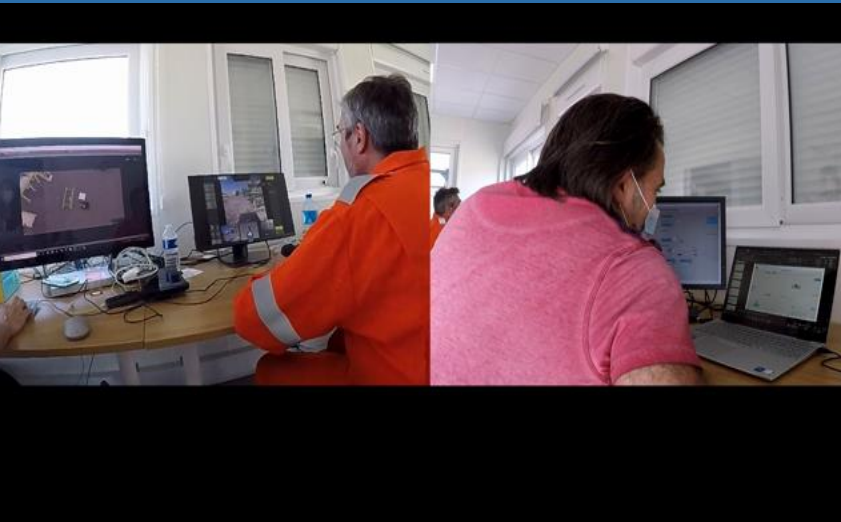
Operation room :
Digital twin (left)
robot remote controller (right)



Control room :
Process controller (powerpointware left)
Work permits (right top)
Process Description (right bottom)

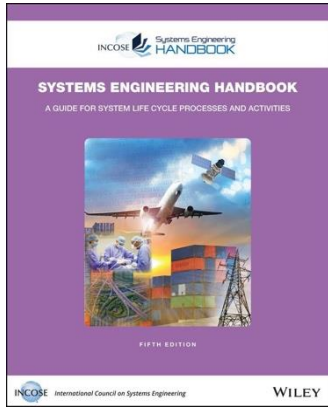


Human in the loop analysis



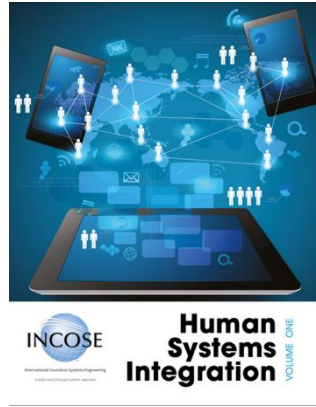
- **Interaction analysis : Type and direction of the interaction.**
 - **Efficiency (time)**
 - **Efficacy (repetition)**
 - **Feedback**
- **User assessment**
 - **Situation awareness evaluation : SART** (Taylor, 1990)
 - Individual Cognitive task load : **NASA-TLX** (Hart 1988)
 - Group Cognitive task load : **TWLQ** (Sellers, 2013)
 - User engagement (Shah 2010)
 - User interface **utilisabilité de l'interface: SUS**
- **BPMN Modelisation**

Additional Resources



2 min

INCOSE Systems Engineering Handbook



2 hour

INCOSE HSI Wg – Human System Integration Primer Vol. 1



2 week

Nasa – Human System Integration Handbook

המכללה
האקדמית להנדסה
בראודה
ביתר עתיד



ESTIA
INSTITUTE OF TECHNOLOGY
une école
CCI BAYONNE
PAYS BASQUE
Ecole d'Ingenierie



2 month

Braude University - Human System Integration Mooc

Estia HSI Course

About the INCOSE HSI Working Group



Working Group

HUMAN SYSTEMS INTEGRATION



HSI WG

Mission and Aims

- Our mission is to ensure all design and engineering disciplines include human and socio-technical interaction and integration to enable system performance.
- The HSI WG aims to demonstrate the value of HSI as an integral part of the Systems Engineering effort and to develop associated practices, resources, organizations, and training to support the Systems Engineering community's adoption of HSI by:
 - Staying at the forefront of evolving SE practices to ensure that the presence of humans is integrated into the engineering in order to deliver successful sociotechnical systems across their life cycle.
 - Fostering a community of practitioners from the contributing disciplines (System Engineering, Human Factors/Ergonomics, Information Technology, and Operational Domains) and develop a common framework.

INCOSE HSI WG

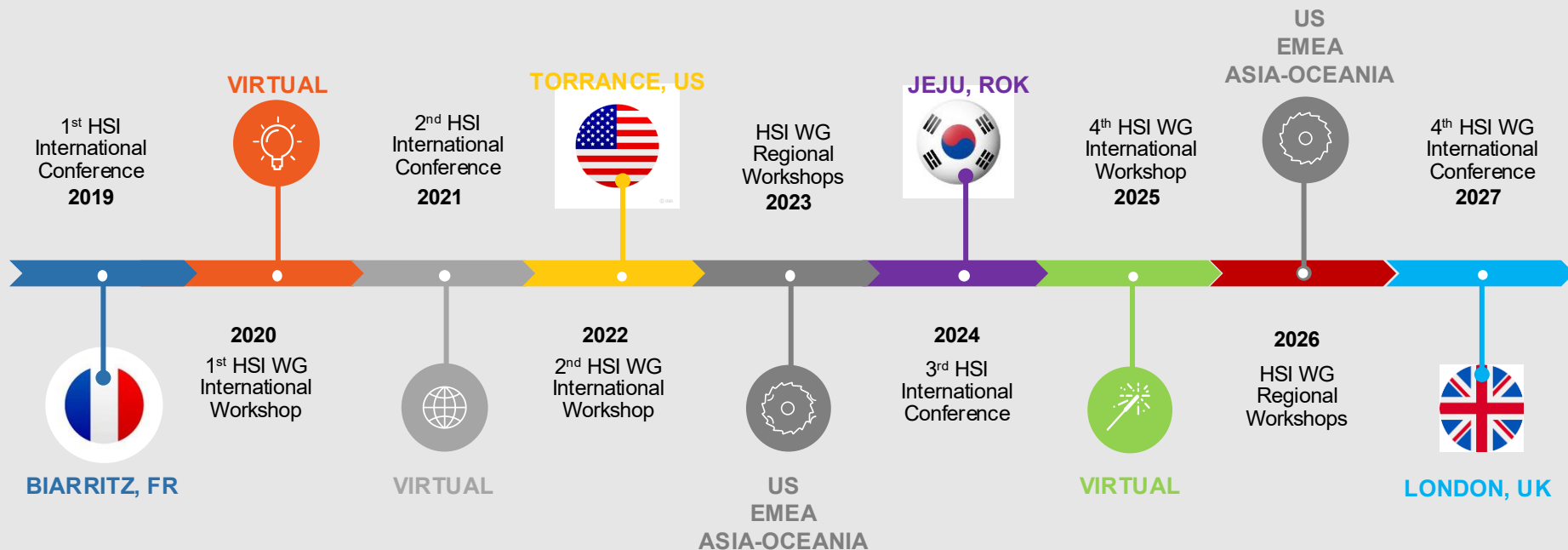
- ~500 members from various INCOSE chapters worldwide
- Academia, Government and Industry
- Covering aerospace, space, defence, automotive, academia, telecoms, transportation, infrastructure



What does the HSI WG do?



INCOSE HSI WG Events



Coming up...

HSI WG General Meetings 3rd Wednesday of every other month (July, Sept, Nov)



Let's connect



Email us on:

hsi@incose.net

Follow our HSI WG Linked In Page:



<https://www.linkedin.com/company/incose-hsi>

Join the HSI Community:



<https://connect.incose.org/home>

Meeting Recordings:



**[http://youtube.com/
@INCOSE_HSI_WG](http://youtube.com/@INCOSE_HSI_WG)**

Thank you